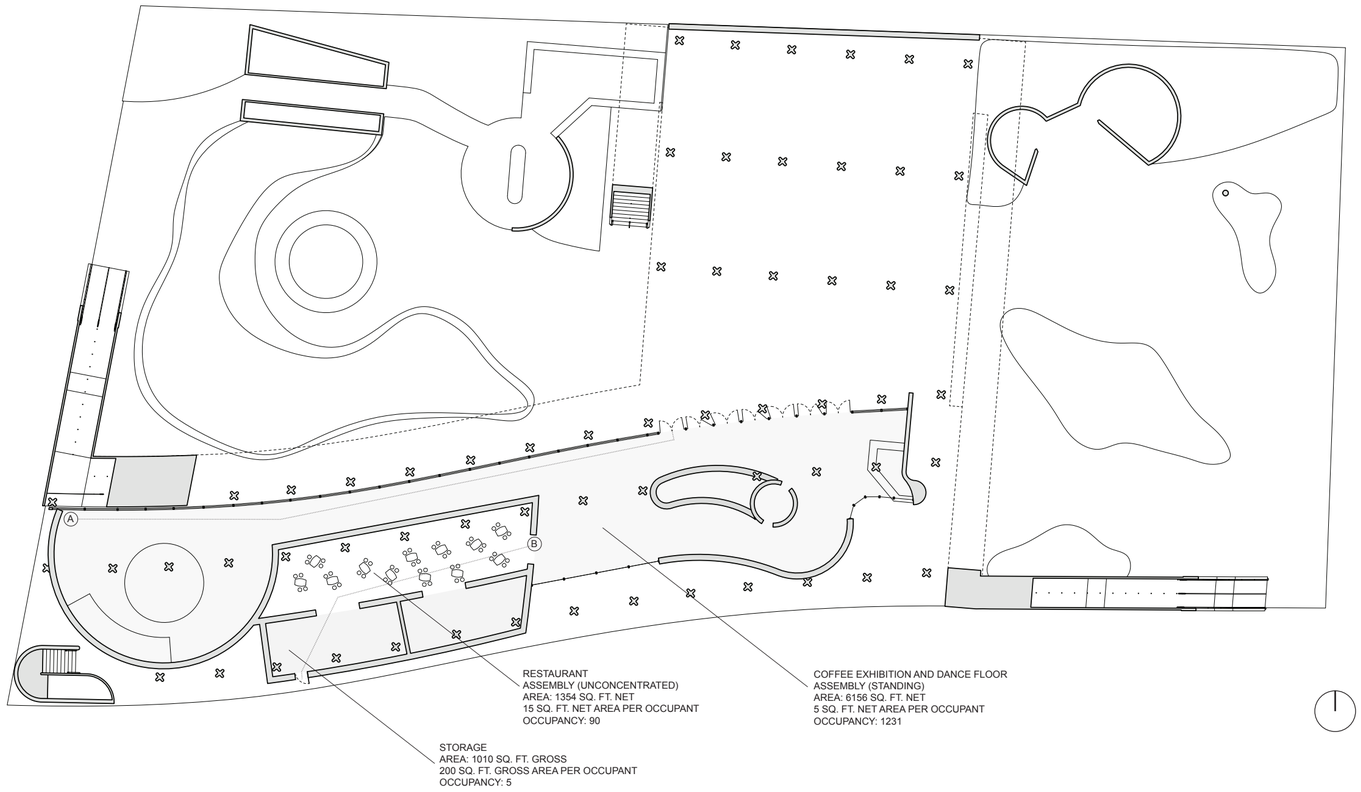
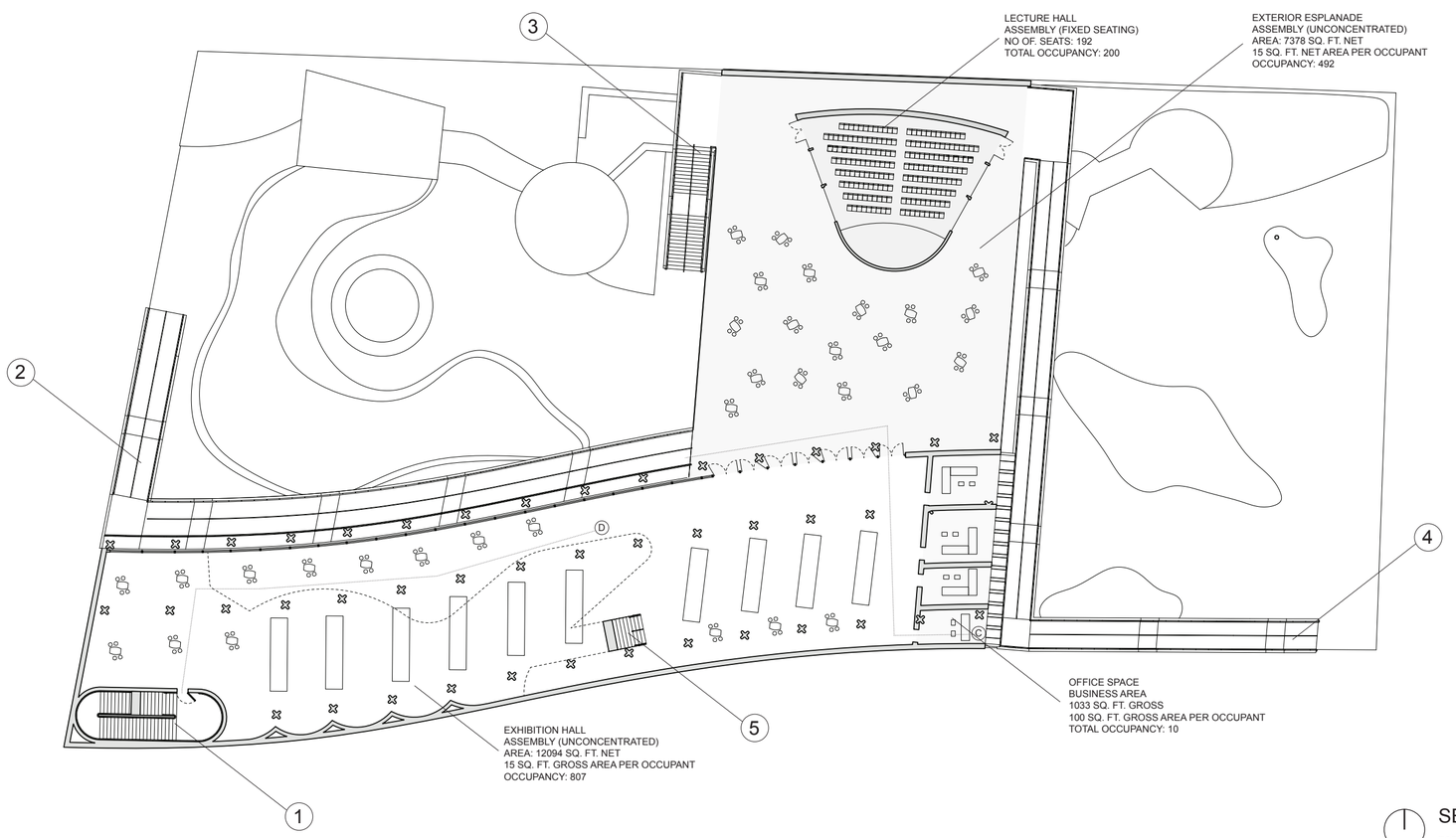


GROUND FLOOR

TOTAL ENCLOSED AREA: 8520 SQ. FT.
TOTAL OCCUPANCY: 1326

SCALE: 1/32" = 1'-0"



SECOND FLOOR

TOTAL AREA: 20705 SQ. FT.
TOTAL OCCUPANCY: 1509*

SCALE: 1/32" = 1'-0"

BUILDING CODE SUMMARY

IBC TABLE 503:
BUILDING OF TYPE 1B CONSTRUCTION WITH OCCUPANCY CLASS A-3 IS ALLOWED AN UNLIMITED FLOOR AREA UP TO 11 STORIES. ORIGINAL BUILDING AT 3 STORIES, COMPLIANT WITH REQUIREMENTS.

IBC TABLE 601:
TYPE 1B CONSTRUCTION REQUIRES A 2 HOUR FIRE RATING. THIS INCREASES TO 3 HOURS UPON ADDITION OF AUTOMATED SPRINKLER SYSTEM COMPLIANT WITH IBC SECTION 903, CHANGING BUILDING TO TYPE 1A CONSTRUCTION.

IBC TABLE 1004.1:
GROUND FLOOR: 1326 OCCUPANTS
SECOND FLOOR: 1509 OCCUPANTS
THIRD FLOOR: 352 OCCUPANTS

IBC TABLE 1005.1:
GROUND FLOOR:
1326 * 0.15 = 199 IN DOOR WIDTH
SECOND FLOOR:
817 * 0.15 = 123 IN DOOR WIDTH (EXHIBITION)
TOTAL PROVIDED: 429 IN
200 * 0.15 = 30 IN DOOR WIDTH (LECTURE)
TOTAL PROVIDED: 134 IN
1685 * 0.2 = 337 IN STAIR WIDTH (TOTAL)
RAMP 1: 96 IN
RAMP 2: 86 IN
STAIR 1: 60 IN
STAIR 2: 96 IN
TOTAL PROVIDED: 338 IN
THIRD FLOOR:
352 * 0.2 = 71 IN STAIR WIDTH
STAIR 1: 60 IN
STAIR 3: 90 IN
TOTAL PROVIDED: 150 IN

SECTION 1005.5 STATES THAT IN CASE OF MORE THAN ONE EXIT, LOSS OF ONE EXIT SHALL NOT REDUCE EGRESS CAPACITY TO LESS THAN 50% OF REQUIRED.

IBC TABLE 1019.1:
OCCUPANT LOAD OF MAIN FLOOR IS GREATER THAN 1000, REQUIRING 4 MEANS OF EGRESS. AS THE MAIN FLOOR AREA, EXCLUDING MEZZANINE, IS ACCESSIBLE, MORE THAN 1 ACCESSIBLE MEANS OF EGRESS IS ALSO REQUIRED.

MEANS OF EGRESS (STAIRS & RAMPS)

ASSIGNMENT 2
PAGE: 1 OF 4

48-215 MATERIALS & ASSEMBLY
SPRING 2017

INSTRUCTOR: DAMIANI

DAN CASCAVAL